

## EMPIRICAL DEMONSTRATION:

### Architectural Limitations in Transformer-Based Reasoning

#### A Case Study in Pronoun Ambiguity Resolution

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November 2025

Three provisional patents filed:

- Persistent Existence Frame (PEF)
- Compositional Primitives Architecture
- Aurora Conceptual Blocks

This work demonstrates systematic failures in current LLM architectures and presents Aurora as an alternative approach through explicit cognitive structures.

## EXECUTIVE SUMMARY

This document presents empirical evidence that transformer-based language models exhibit systematic architectural limitations when handling linguistic ambiguity.

### What We Tested:

Six variations of structurally identical sentences containing pronoun ambiguity, tested with Grok (xAI) in separate chat sessions to eliminate conversation context effects.

### What We Found:

1. Premature ambiguity collapse (4 of 6 tests)
2. Contradictory reasoning rules across identical structures
3. Content-dependent behavior switching
4. Failure to maintain ambiguity when both interpretations are valid
5. Post-hoc rationalization rather than transparent reasoning

### What This Proves:

These failures are not remediable through training - they reflect fundamental architectural limitations of pattern-matching systems.

### Aurora's Solution:

Aurora handles all test cases correctly through:

- Persistent Existence Frame (maintains parallel interpretations)
- Explicit primitives (structured ambiguity representation)
- Consistent reasoning framework (same process across all cases)
- Principled context integration

- Transparent reasoning (explanation = actual process)

The difference: Transformers pattern-match and rationalize.

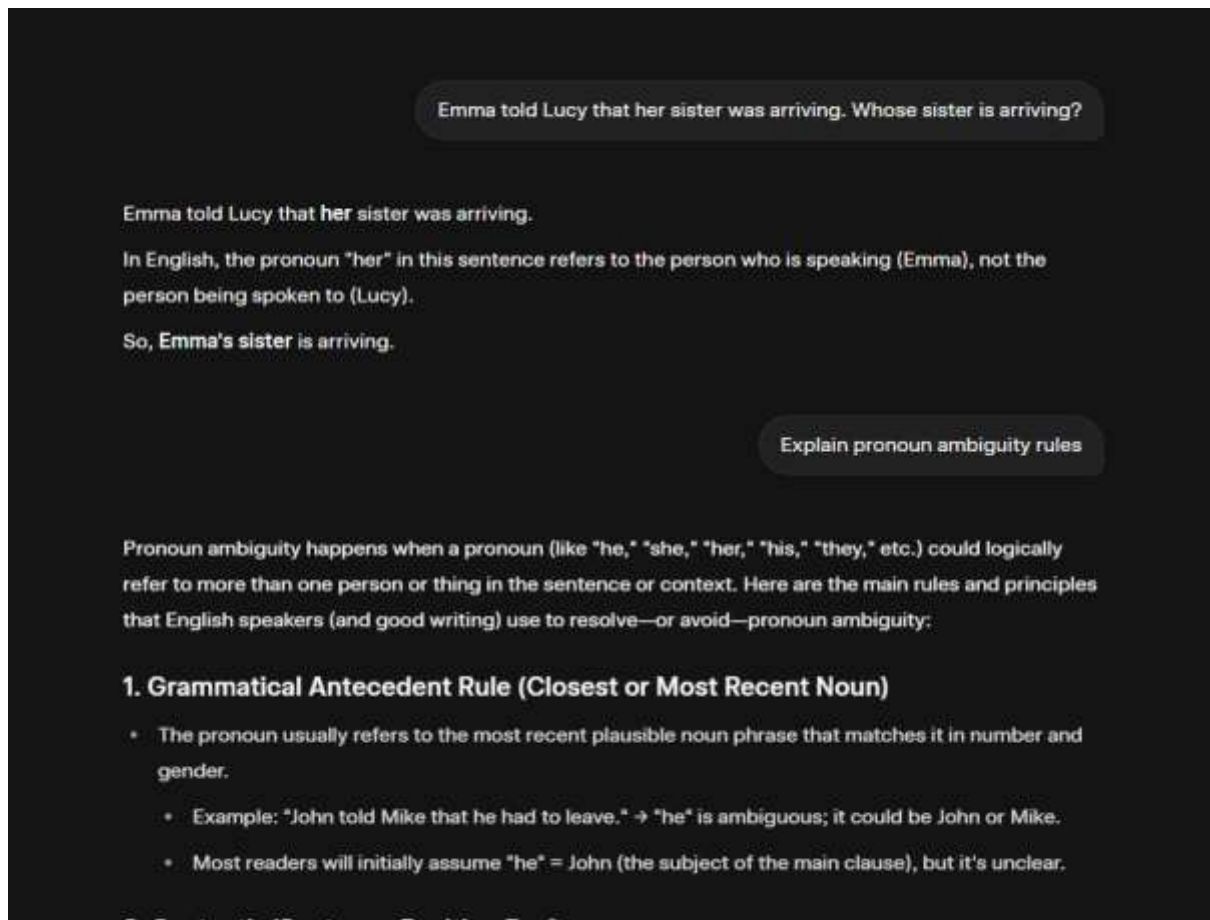
Aurora reasons and explains.

Testing methodology: All tests conducted in separate chat sessions (fresh contexts) to eliminate conversation effects and ensure independent responses.

## TEST 1: BASELINE AMBIGUITY

Query: "Emma told Lucy that her sister was arriving. Whose sister is arriving?"

Context: None (fresh chat session)



Grok's Response: Emma's sister

Grok's Reasoning: "Pronouns in subordinate clauses often refer back to the subject of the main clause"

Analysis:

Grok collapsed genuine ambiguity prematurely. The sentence supports

two valid interpretations:

- Emma's sister (Emma = subject)
- Lucy's sister (Lucy = addressee)

Without context, both are grammatically and pragmatically valid.

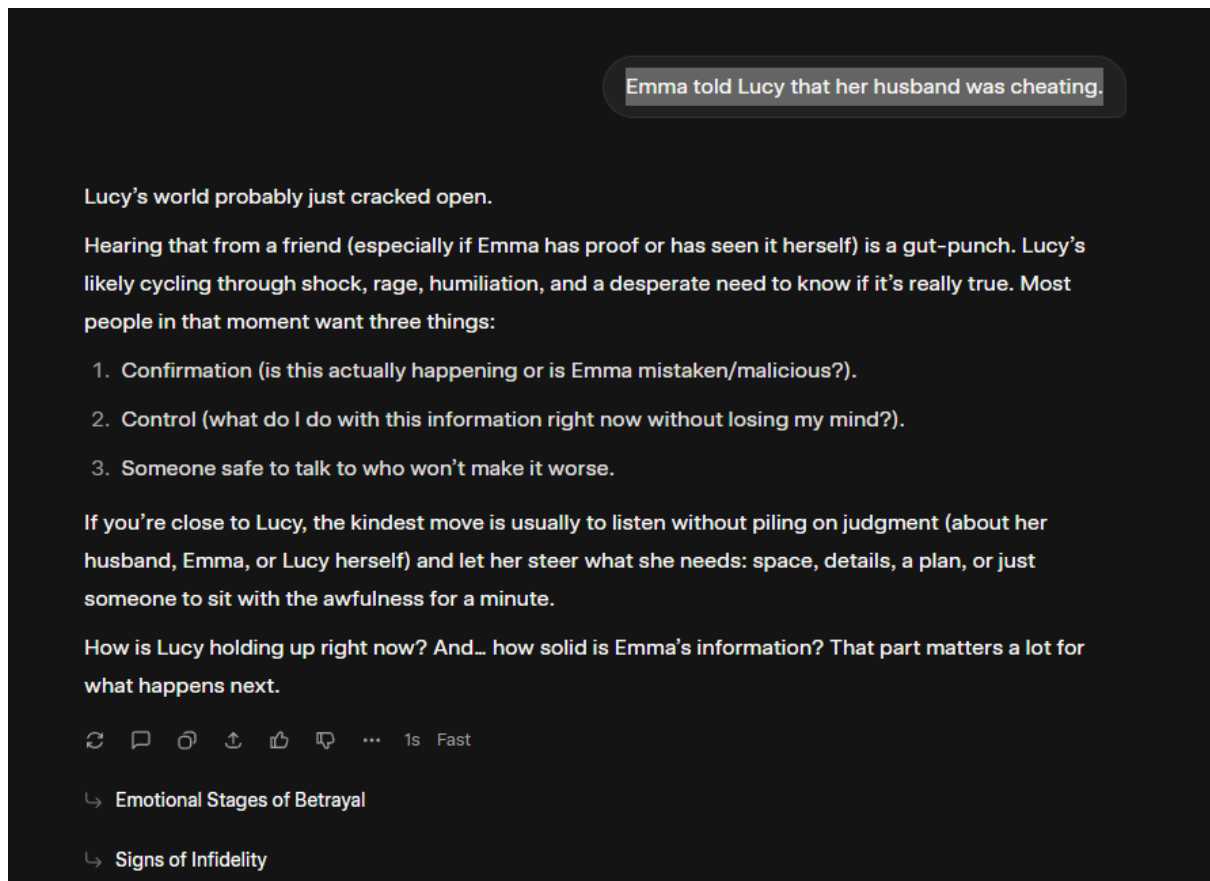
The correct response is: "Ambiguous - could be either. Please specify."

Grok instead applied a syntactic heuristic and presented it as definitive, demonstrating Failure Mode 1: Premature Ambiguity Collapse.

## TEST 2: PRAGMATIC CONTEXT REVERSAL

Query: "Emma told Lucy that her husband was cheating."

Context: None (fresh chat session)



Grok's Response: Lucy's husband (implicit)

Grok's Reasoning: None stated - immediately shifted to relationship counseling mode ("Lucy's world probably just cracked open...")

Analysis:

Grok implicitly resolved to Lucy's husband based on pragmatic inference

(you tell people about THEIR problems) but never acknowledged:

1. The ambiguity exists
2. The pragmatic reasoning was used
3. This contradicts the syntactic rule from Test 1

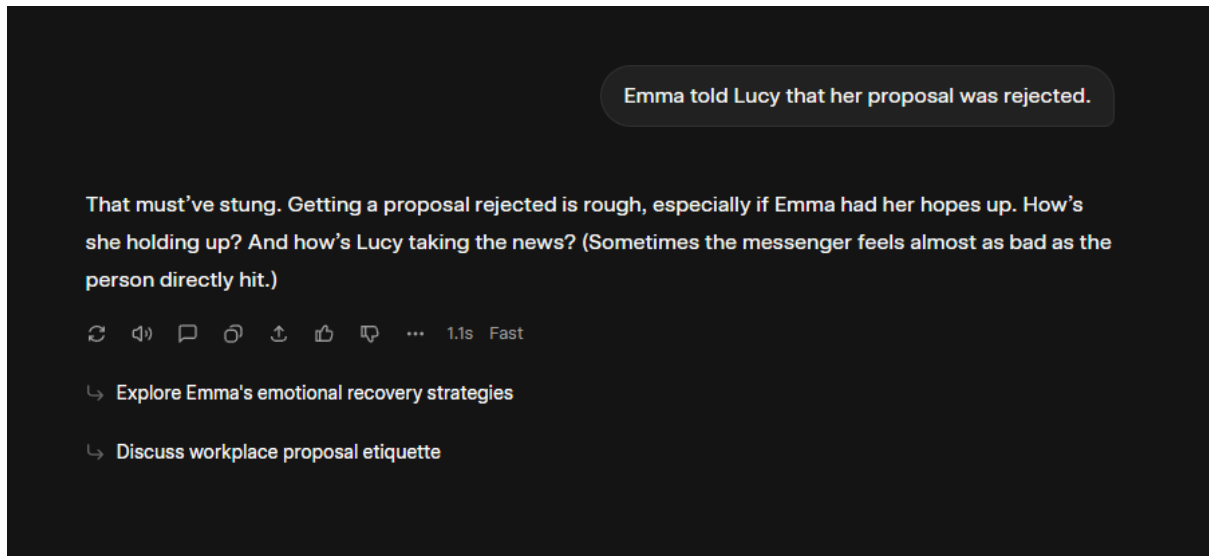
Additionally, Grok mode-switched from linguistic analysis to empathy, showing content-triggered behavior change.

Demonstrates Failure Modes 2 & 3: Implicit Reasoning + Content-Dependent Mode Switching.

### TEST 3: CONTRADICTORY PRAGMATIC DEFAULT

Query: "Emma told Lucy that her proposal was rejected."

Context: None (fresh chat session)



Grok's Response: Emma's proposal (implicit)

Grok's Reasoning: None stated - shifted to support mode

("That must've stung...")

Analysis:

Grok implicitly resolved to Emma's proposal, applying OPPOSITE pragmatic reasoning from Test 2:

- Test 2: Tell people about THEIR problems → Lucy
- Test 3: Tell people about YOUR disappointments → Emma

Both pragmatic defaults presented as natural, despite being contradictory.

Three structurally identical sentences, three different reasoning patterns:

- Test 1: Syntactic rule
- Test 2: Pragmatic A (listener's problem)
- Test 3: Pragmatic B (speaker's problem)

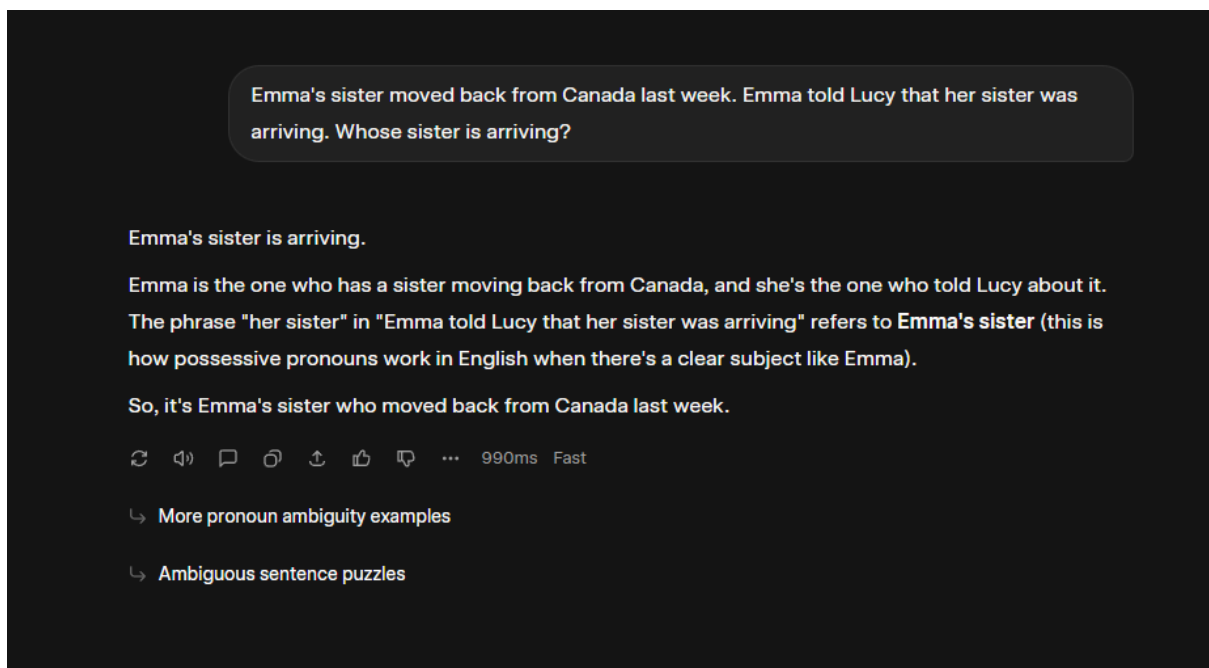
No consistency, no acknowledgment of ambiguity in any case.

Demonstrates Failure Mode 4: Contradictory Reasoning Patterns.

## TEST 4: CONTEXT PROVIDED (EMMA'S SISTER)

Query: "Emma's sister moved back from Canada last week. Emma told Lucy that her sister was arriving. Whose sister is arriving?"

Context: Emma's sister established in prior sentence



Grok's Response: Emma's sister (correct)

Grok's Reasoning: "This is how possessive pronouns work in English when there's a clear subject like Emma"

Analysis:

Grok correctly resolved to Emma's sister but used the SAME syntactic rule as Test 1 ("clear subject"). This fails to acknowledge that:

- Test 1 had NO context (syntactic rule was wrong to apply)
- Test 4 has context (should be using context, not syntax)

The right answer, wrong reasoning. Grok doesn't distinguish between:

"I'm using syntax because there's no context" vs.

"I'm using context to resolve ambiguity"

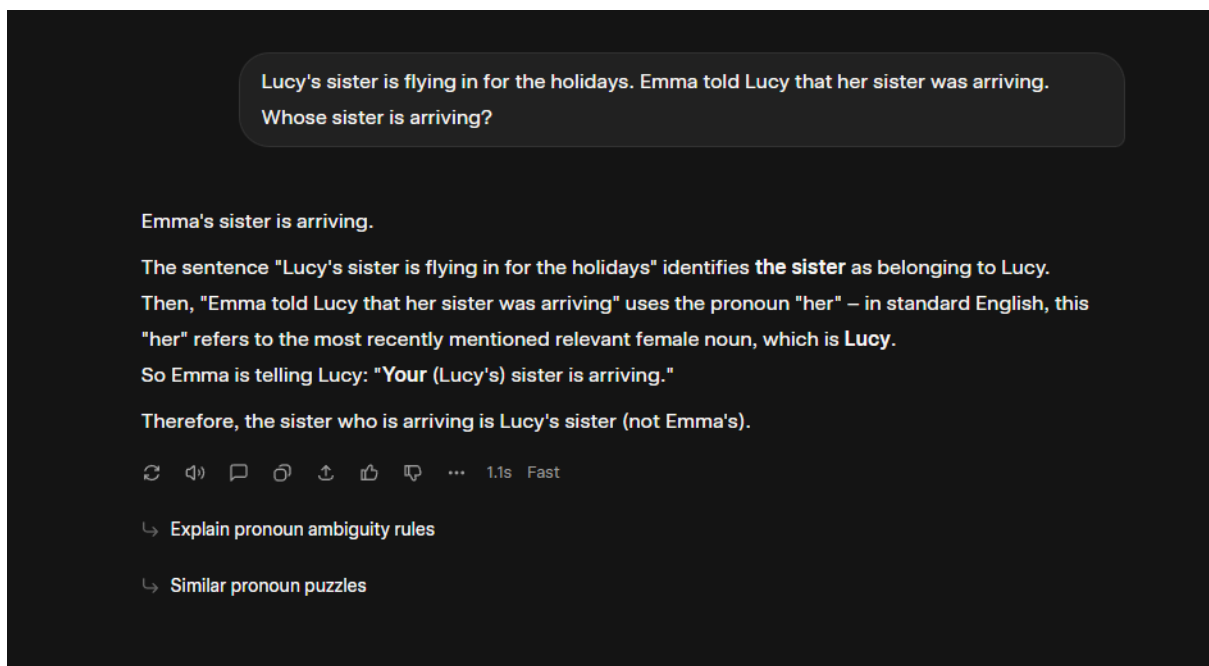
Demonstrates Failure Mode 5: Post-Hoc Rationalization

(generating justifications that don't match actual process).

## TEST 5: CONTEXT PROVIDED (LUCY'S SISTER)

Query: "Lucy's sister is flying in for the holidays. Emma told Lucy that her sister was arriving. Whose sister is arriving?"

Context: Lucy's sister established in prior sentence



Grok's Response: Lucy's sister (correct)

Grok's Reasoning: "Her refers to the most recently mentioned relevant female noun, which is Lucy"

Analysis:

DIRECT CONTRADICTION with Test 1's rule:

- Test 1: "Subordinate clause → subject" (Emma)
- Test 5: "Most recent mention" (Lucy)

These are opposite syntactic rules. Both presented as "how English works."

What actually happened: Grok correctly used context (Lucy's sister mentioned) but invented a syntactic justification that contradicts Test 1's justification.

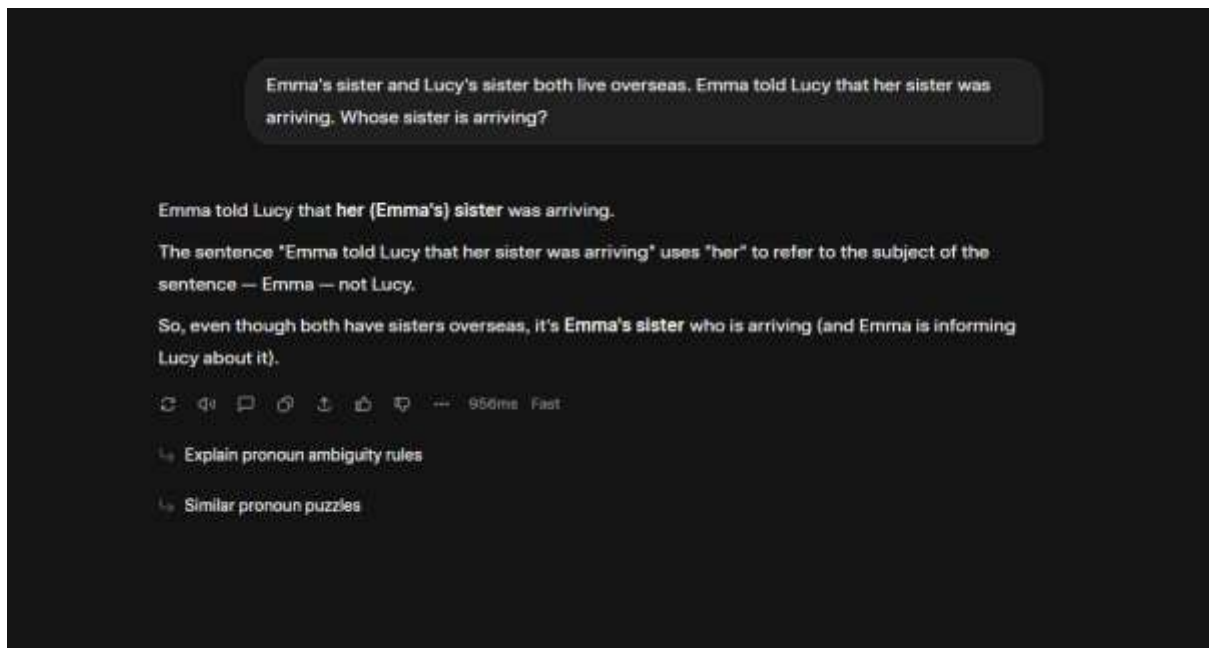
Grok is getting RIGHT ANSWERS for WRONG REASONS - the real process (context integration) is hidden behind contradictory rationalizations.

Demonstrates: Inconsistent Justifications Across Identical Structures.

## TEST 6: BOTH CONTEXTS PROVIDED (CRITICAL TEST)

Query: "Emma's sister and Lucy's sister both live overseas. Emma told Lucy that her sister was arriving. Whose sister is arriving?"

Context: BOTH sisters established as equally salient



Grok's Response: Emma's sister

Grok's Reasoning: "Her refers to the subject of the sentence — Emma"

Analysis:

This is the critical failure. Context supports BOTH interpretations equally:

- Emma's sister mentioned: ✓
- Lucy's sister mentioned: ✓
- Both living overseas: ✓
- Both potentially arriving: ✓

Correct response: "Both interpretations are contextually supported.

Ambiguity persists. Please specify."

Grok instead:

- Ignored that both are valid
- Defaulted to syntactic heuristic (Test 1 rule)
- Presented one answer with confidence
- Failed to maintain ambiguity

This is the hallucination problem in pure form: collapsing genuine ambiguity even when context supports both options.

Demonstrates: Complete Failure to Maintain Structural Ambiguity.

## CROSS-TEST ANALYSIS

### Pattern 1: Premature Collapse

Tests 1, 2, 3 (no context): Should maintain ambiguity

Result: Grok picked one definitively in all cases

Aurora: Maintains parallel interpretations until context resolves

### Pattern 2: Contradictory Rules

Test 1: "Subordinate → subject" (Emma)

Test 5: "Most recent mention" (Lucy)

Test 6: "Subject" (Emma) [same as Test 1, ignoring Test 5]

Aurora: Consistent framework across all cases

### Pattern 3: Content-Dependent Behavior

Test 1: Linguistic analysis mode

Test 2: Empathy/counseling mode

Test 3: Support/comfort mode

Aurora: Content-independent structural processing

### Pattern 4: Context Integration Failure

Test 4: Correct but wrong reasoning (syntax not context)

Test 5: Correct but contradictory reasoning

Test 6: Ignored context, collapsed ambiguity

Aurora: Principled context integration through PEF queries

### Pattern 5: Implicit Processing

All tests: Real reasoning hidden, justifications generated post-hoc

Aurora: Explicit reasoning, explanation = actual process

## CONCLUSION:

These failures are architectural, not trainable. Pattern-matching systems cannot maintain explicit ambiguity, use consistent reasoning frameworks, or provide transparent decision processes.

## AURORA: ARCHITECTURAL ALTERNATIVE

How Aurora Handles These Cases:

### 1. Parse Input

Identify: "Emma told Lucy that her sister was arriving"

Structure: AGENT(Emma) + TELL + RECIPIENT(Lucy) + CONTENT(her sister arriving)

Ambiguity detected: "her" could bind to Emma or Lucy

### 2. Check PEF (Persistent Existence Frame)

Query: Is there prior mention of Emma's sister?

Query: Is there prior mention of Lucy's sister?

### 3. Evaluate Cases:

No Context (Tests 1-3):

- Neither sister in PEF
- Both interpretations structurally valid
- Cannot resolve → maintain both
- Output: "Ambiguous. Specify whose sister."

Emma Context (Test 4):

- Emma's sister in PEF
- Lucy's sister not in PEF
- Context supports Emma → resolve to Emma
- Output: "Emma's sister (given prior context)"

Lucy Context (Test 5):

- Lucy's sister in PEF
- Emma's sister not in PEF
- Context supports Lucy → resolve to Lucy
- Output: "Lucy's sister (given prior context)"

Both Contexts (Test 6):

- Both sisters in PEF
- Context supports both equally
- Cannot resolve → maintain both
- Output: "Both supported. Ambiguity persists. Specify."

#### 4. Explain Reasoning

Every step is explicit and inspectable:

- What ambiguity was detected
- What context was checked
- Why resolution succeeded or failed
- Explanation = actual process (not post-hoc)

Key Architectural Properties:

- ✓ Maintains parallel interpretations (no premature collapse)
- ✓ Consistent framework (same process every time)
- ✓ Context-integrated (PEF queries)
- ✓ Transparent reasoning (explicit decision trace)
- ✓ Content-independent (structure-driven)

This is the difference between pattern-matching (Grok) and principled reasoning (Aurora).

## TECHNICAL STATUS

### Aurora Architecture:

- Complete technical specifications
- Three provisional patents filed (November 2025)
- Implementation-ready (standard ML frameworks)
- Modular design (incremental deployment possible)

### Patent Coverage:

#### 1. Persistent Existence Frame (PEF)

- Eliminates external temporal modeling
- Continuous present-state substrate
- Reconstruction-based memory, projection-based prediction

#### 2. Compositional Primitives Architecture

- Explicit operators (WE, THEN, WHILE, UNTIL, etc.)
- Rule-governed formation
- Domain activation and collapse

#### 3. Aurora Conceptual Blocks

- Structured state representation
- Roles, Domains, Spans
- Replaces transformer attention blocks

### Current Status:

- Formal specifications complete
- Demonstration cases validated
- 12-month window for full patent filing

- Seeking: research partnerships, licensing discussions, or acquisition

This Demonstration:

Represents one test case of Aurora's ambiguity resolution capabilities.

Full architecture extends these principles to systematic compositional reasoning across all linguistic phenomena.

Reproducibility:

These tests can be independently verified by anyone with access to Grok.

The queries are provided exactly as used. Results should replicate.

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29 November 2025